

# PN7221 RF参数合入

更新配置文件libnfc-nxp-rfExt.conf

操作步骤:

1. 安装NXP NFC Cockpit v8.3.0.0, 安装过程略。安装包: NxpNfcCockpit\_08.03.00.zip

2. 天线厂提供XML配置文件, 如: 5P7V\_5611.xml

3. 连接终端设备, 执行脚本

C:\nxp\NxpNfcCockpit\_v8.3.0.0\firmware\Secondary\_Pn722x\Android\PN722x\_UcBalAndroid\_LogEnabled.bat, 注意查看执行过程是否有报错

```
Waiting for devices
restarting adbd as root
Verity is already disabled
Remounted /system as RW
Remounted /system_ext as RW
Remounted /vendor as RW
Remounted /vendor_dlkm as RW
Remounted /system_dlkm as RW
Remounted /odm as RW
Remounted /product as RW
Remount succeeded
pnsr_nfc: 1 file pushed, 0 skipped. 86.6 MB/s (92384 bytes in 0.001s)

*****
Steps to start the server
*****
Perform the following steps to start Android MW Server
1. cd /system/bin
2. pnsr true/false (true : Enable Debug info )
                   (false : Disable Debug info )
3. This will start the server and keep waiting for client
*****
*****

*****
Mandatory!
Step to restore the original Android Middleware HAL (NFC HAL)
*****
Once you are done with Cockpit tool.
1. Kill the pnsr by pressing Ctrl+C
2. run restore_NFC_HAL.bat (to restore the NFC service)
*****

adbd is already running as root
Verity is already disabled
Remounted /system as RW
Remounted /system_ext as RW
Remounted /vendor as RW
Remounted /vendor_dlkm as RW
Remounted /system_dlkm as RW
Remounted /odm as RW
Remounted /product as RW
Remount succeeded
MC-E1o-SM6115-8G-NFC:/ # _
```

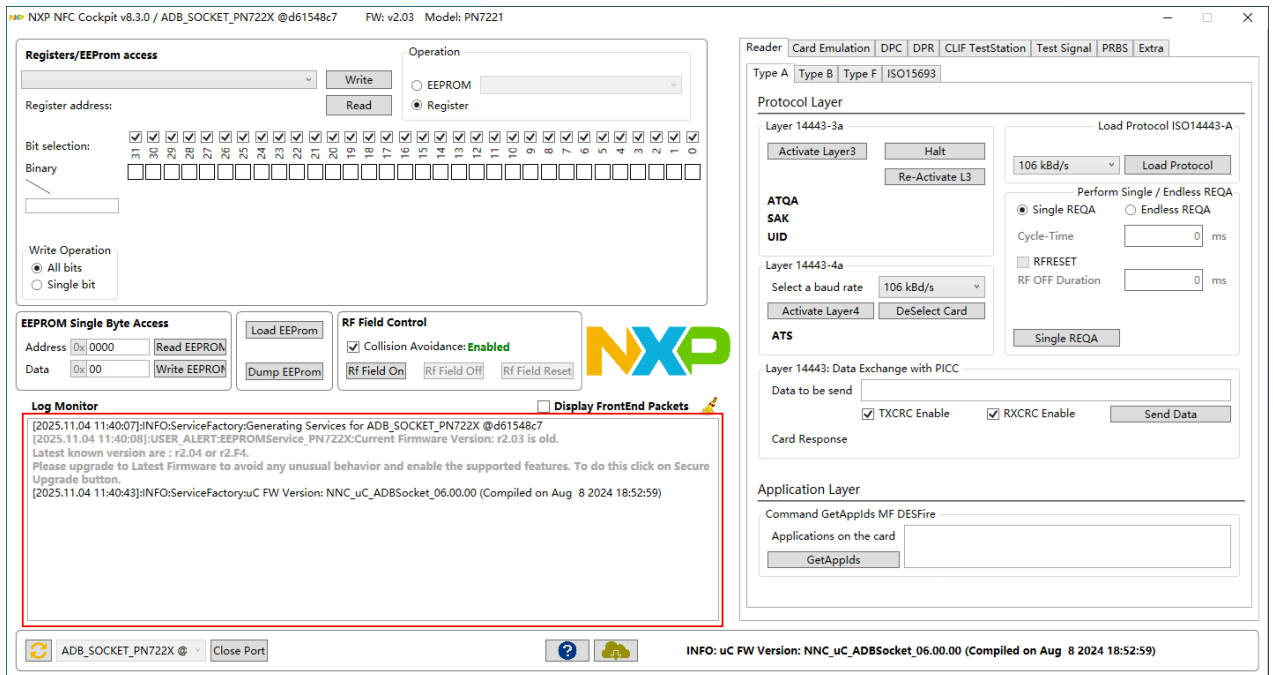
4. 执行指令pnsr true开启8059端口监听

```
MC-E1o-SM6115-8G-NFC:/ # pnsr true
***** NXP Semiconductor *****

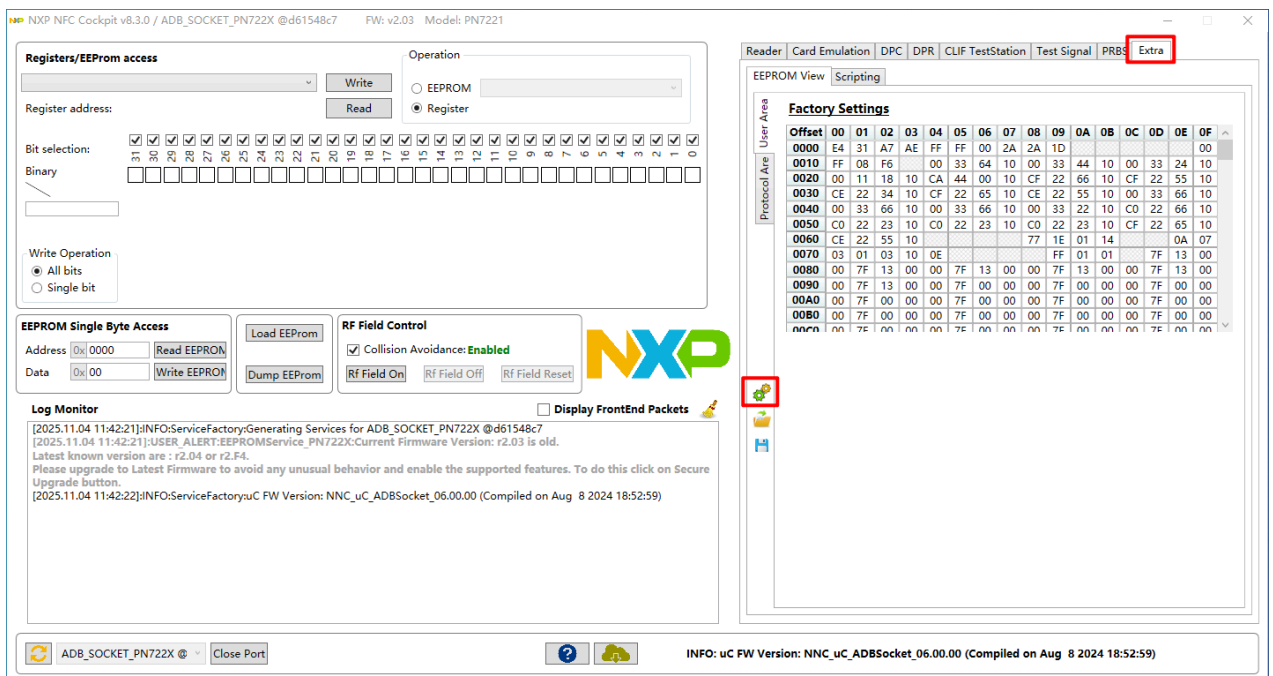
NNC_uC_ADBSocket_06.00.00

*****
12:07:08.439 SEND : 4 > 20000100
12:07:08.467 RCEIVE : 4 < 40000100
12:07:08.470 RCEIVE : 13 < 60000A02002004052153030203
12:07:08.472 SEND : 5 > 2001020000
12:07:08.476 RCEIVE : 23 < 40011400000E000001C000FF0000FF0003010002008000
12:07:08.479 SEND : 4 > 2F000100
12:07:08.482 RCEIVE : 4 < 4F000100
Socket successfully created..
Listening at port number 8059
Socket successfully binded
Server listening..
```

## 5. 打开工具NXP NFC Cockpit v8.3.0.0



## 6. 找到Extra



## 7. 导入XML配置文件

Registers/EEPROM access

Register address:

Write

Read

Operation

EEPROM

Register

Bit selection:

15

14

13

12

11

10

9

8

7

6

5

4

3

2

1

0

Binary

Write Operation

All bits

Single bit

EEPROM Single Byte Access

Address: 0x0000

Read EEPROM

Data: 0x00

Write EEPROM

Load EEPROM

Dump EEPROM

RF Field Control

Collision Avoidance: Enabled

RF Field On

RF Field Off

RF Field Reset

Log Monitor

Display FrontEnd Packets

[2025.11.04 11:42:21]:INFO:ServiceFactory:Generating Services for ADB\_SOCKET\_PN722X @d61548c7

[2025.11.04 11:42:21]:USER\_ALERT:EEPROMService\_PN722X:Current Firmware Version: r2.03 is old.

Latest known version are : r2.04 or r2.F4.

Please upgrade to Latest Firmware to avoid any unusual behavior and enable the supported features. To do this click on Secure Upgrade button.

[2025.11.04 11:42:22]:INFO:ServiceFactory:uC FW Version: NNC\_uC\_ADBSocket\_06.00.00 (Compiled on Aug 8 2024 18:52:59)

Reader

Card Emulation

DPC

DPR

CLIF TestStation

Test Signal

PRBS

Extra

EEPROM View

Scripting

Factory Settings

| Offset | 00 | 01 | 02 | 03 | 04 | 05 | 06 | 07 | 08 | 09 | 0A | 0B | 0C | 0D | 0E | 0F |
|--------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| 0000   | E4 | 31 | A7 | AE | FF | FF | 00 | 2A | 2A | 1D |    |    |    |    |    | 00 |
| 0010   | FF | 08 | F6 |    | 00 | 33 | 64 | 10 | 00 | 33 | 44 | 10 | 00 | 33 | 24 | 10 |
| 0020   | 00 | 11 | 18 | 10 | CA | 44 | 00 | 10 | CF | 22 | 66 | 10 | CF | 22 | 55 | 10 |
| 0030   | CE | 22 | 34 | 10 | CF | 22 | 65 | 10 | CE | 22 | 55 | 10 | 00 | 33 | 66 | 10 |
| 0040   | 00 | 33 | 66 | 10 | 00 | 33 | 66 | 10 | 00 | 33 | 22 | 10 | C0 | 22 | 66 | 10 |
| 0050   | C0 | 22 | 23 | 10 | C0 | 22 | 23 | 10 | C0 | 22 | 23 | 10 | CF | 22 | 65 | 10 |
| 0060   | CE | 22 | 55 | 10 |    |    |    |    | 77 | 1E | 01 | 14 |    |    | 0A | 07 |
| 0070   | 03 | 01 | 03 | 10 | 0E |    |    |    | FF | 01 | 01 |    |    | 7F | 13 | 00 |
| 0080   | 00 | 7F | 13 | 00 | 00 | 7F | 13 | 00 | 00 | 7F | 13 | 00 | 00 | 7F | 13 | 00 |
| 0090   | 00 | 7F | 13 | 00 | 00 | 7F | 00 | 00 | 00 | 7F | 00 | 00 | 00 | 7F | 00 | 00 |
| 00A0   | 00 | 7F | 00 | 00 | 00 | 7F | 00 | 00 | 00 | 7F | 00 | 00 | 00 | 7F | 00 | 00 |
| 00B0   | 00 | 7F | 00 | 00 | 00 | 7F | 00 | 00 | 00 | 7F | 00 | 00 | 00 | 7F | 00 | 00 |
| 00C0   | nn | nn | nn | nn | nn | nn | nn | nn | nn | nn | nn | nn | nn | nn | nn | nn |

Hardware Settings

| Offset | 00 | 01 | 02 | 03 | 04 | 05 | 06 | 07 | 08 | 09 | 0A | 0B | 0C | 0D | 0E | 0F |
|--------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
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| 0040   | 00 | 33 | 66 | 10 | 00 | 33 | 66 | 10 | 00 | 33 | 22 | 10 | C0 | 22 | 66 | 10 |
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| 0060   | CE | 22 | 55 | 10 |    |    |    |    | 77 | 4A | 01 | 14 |    |    | 0A | 07 |
| 0070   | 03 | 01 | 03 | 10 | 0E |    |    |    | FF | 01 | 01 |    |    | 28 | 13 | 00 |
| 0080   | 00 | 28 | 13 | 00 | 00 | 28 | 13 | 00 | 00 | 26 | 13 | 00 | 00 | 23 | 13 | 00 |
| 0090   | 00 | 21 | 13 | 00 | 00 | 20 | 00 | 00 | 00 | 1F | 00 | 00 | 00 | 1E | 00 | 00 |
| 00A0   | 00 | 1D | 00 | 00 | 00 | 1D | 00 | 00 | 00 | 1D | 00 | 00 | 00 | 1C | 00 | 00 |
| 00B0   | 00 | 1C | 00 | 00 | 00 | 1B | 00 | 00 | 00 | 1B | 00 | 00 | 00 | 1A | 00 | 00 |
| 00C0   | nn | nn | nn | nn | nn | nn | nn | nn | nn | nn | nn | nn | nn | nn | nn | nn |

ADB\_SOCKET\_PN722X

Close Port

INFO: uC FW Version: NNC\_uC\_ADBSocket\_06.00.00 (Compiled on Aug 8 2024 18:52:59)

## 8. 保存libnfc-nxp-rfExt.conf配置文件

Registers/EEPROM access

Register address:

Write

Read

Operation

EEPROM

Register

Bit selection:

15

14

13

12

11

10

9

8

7

6

5

4

3

2

1

0

Binary

Write Operation

All bits

Single bit

EEPROM Single Byte Access

Address: 0x0000

Read EEPROM

Data: 0x00

Write EEPROM

Load EEPROM

Dump EEPROM

RF Field Control

Collision Avoidance: Enabled

RF Field On

RF Field Off

RF Field Reset

Log Monitor

Display FrontEnd Packets

[2025.11.04 11:42:21]:INFO:ServiceFactory:Generating Services for ADB\_SOCKET\_PN722X @d61548c7

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[2025.11.04 11:42:22]:INFO:ServiceFactory:uC FW Version: NNC\_uC\_ADBSocket\_06.00.00 (Compiled on Aug 8 2024 18:52:59)

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Extra

EEPROM View

Scripting

Factory Settings

| Offset | 00 | 01 | 02 | 03 | 04 | 05 | 06 | 07 | 08 | 09 | 0A | 0B | 0C | 0D | 0E | 0F |
|--------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| 0000   | E4 | 31 | A7 | AE | FF | FF | 00 | 2A | 2A | 1D |    |    |    |    |    | 00 |
| 0010   | FF | 08 | F6 |    | 00 | 33 | 64 | 10 | 00 | 33 | 44 | 10 | 00 | 33 | 24 | 10 |
| 0020   | 00 | 11 | 18 | 10 | CA | 44 | 00 | 10 | CF | 22 | 66 | 10 | CF | 22 | 55 | 10 |
| 0030   | CE | 22 | 34 | 10 | CF | 22 | 65 | 10 | CE | 22 | 55 | 10 | 00 | 33 | 66 | 10 |
| 0040   | 00 | 33 | 66 | 10 | 00 | 33 | 66 | 10 | 00 | 33 | 22 | 10 | C0 | 22 | 66 | 10 |
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| 0060   | CE | 22 | 55 | 10 |    |    |    |    | 77 | 4A | 01 | 14 |    |    | 0A | 07 |
| 0070   | 03 | 01 | 03 | 10 | 0E |    |    |    | FF | 01 | 01 |    |    | 28 | 13 | 00 |
| 0080   | 00 | 28 | 13 | 00 | 00 | 28 | 13 | 00 | 00 | 26 | 13 | 00 | 00 | 23 | 13 | 00 |
| 0090   | 00 | 21 | 13 | 00 | 00 | 20 | 00 | 00 | 00 | 1F | 00 | 00 | 00 | 1E | 00 | 00 |
| 00A0   | 00 | 1D | 00 | 00 | 00 | 1D | 00 | 00 | 00 | 1D | 00 | 00 | 00 | 1C | 00 | 00 |
| 00B0   | 00 | 1C | 00 | 00 | 00 | 1B | 00 | 00 | 00 | 1B | 00 | 00 | 00 | 1A | 00 | 00 |
| 00C0   | nn | nn | nn | nn | nn | nn | nn | nn | nn | nn | nn | nn | nn | nn | nn | nn |

Hardware Settings

| Offset | 00 | 01 | 02 | 03 | 04 | 05 | 06 | 07 | 08 | 09 | 0A | 0B | 0C | 0D | 0E | 0F |
|--------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| 0000   | E4 | 31 | A7 | AE | FF | FF | 00 | 2A | 2A | 1D |    |    |    |    |    | 00 |
| 0010   | FF | 08 | F6 |    | 00 | 33 | 64 | 10 | 00 | 33 | 44 | 10 | 00 | 33 | 24 | 10 |
| 0020   | 00 | 11 | 18 | 10 | CA | 44 | 00 | 10 | CF | 22 | 66 | 10 | CF | 22 | 55 | 10 |
| 0030   | CE | 22 | 34 | 10 | CF | 22 | 65 | 10 | CE | 22 | 55 | 10 | 00 | 33 | 66 | 10 |
| 0040   | 00 | 33 | 66 | 10 | 00 | 33 | 66 | 10 | 00 | 33 | 22 | 10 | C0 | 22 | 66 | 10 |
| 0050   | C0 | 22 | 23 | 10 | C0 | 22 | 23 | 10 | C0 | 22 | 23 | 10 | CF | 22 | 65 | 10 |
| 0060   | CE | 22 | 55 | 10 |    |    |    |    | 77 | 4A | 01 | 14 |    |    | 0A | 07 |
| 0070   | 03 | 01 | 03 | 10 | 0E |    |    |    | FF | 01 | 01 |    |    | 28 | 13 | 00 |
| 0080   | 00 | 28 | 13 | 00 | 00 | 28 | 13 | 00 | 00 | 26 | 13 | 00 | 00 | 23 | 13 | 00 |
| 0090   | 00 | 21 | 13 | 00 | 00 | 20 | 00 | 00 | 00 | 1F | 00 | 00 | 00 | 1E | 00 | 00 |
| 00A0   | 00 | 1D | 00 | 00 | 00 | 1D | 00 | 00 | 00 | 1D | 00 | 00 | 00 | 1C | 00 | 00 |
| 00B0   | 00 | 1C | 00 | 00 | 00 | 1B | 00 | 00 | 00 | 1B | 00 | 00 | 00 | 1A | 00 | 00 |
| 00C0   | nn | nn | nn | nn | nn | nn | nn | nn | nn | nn | nn | nn | nn | nn | nn | nn |

ADB\_SOCKET\_PN722X

Close Port

INFO: uC FW Version: NNC\_uC\_ADBSocket\_06.00.00 (Compiled on Aug 8 2024 18:52:59)

## 9. 配置文件内容如下

```
##### PN7221 libnfc-nxp-rfExt.conf #####

#####

# NXP_NUM_OF_RFEXT_CONFIG : Number of the RF configuration needs to be set.
# MW will read the value of this macro and read the that many config from the
# config file. eg NXP_NUM_OF_RFEXT_CONFIG is 0x02 MW will read NXP_RFEXT_CONFIG_1
# and NXP_RFEXT_CONFIG_2.

NXP_NUM_OF_RFEXT_CONFIG=0x17

#####

# RF-EXT Configuration, Fill the NXP_RFEXT_CONFIG_x with valid NCI pkt.
# eg. NXP_RFEXT_CONFIG_1={20, 02, 0C, 01, A2, 00, 08, E4, 31, A7, AE, FF, FF, 00, 2A}
=> UserArea
```

```
# eg. NXP_RFEXT_CONFIG_2={20, 02, 0A, 01, A0, 0D, 06, 00, 15, 89, 22, 00, 00}
=> ProtocolArea

# UserArea Settings
#####
NXP_RFEXT_CONFIG_1={20, 02, 0C, 01, A2, 0D, 08, 77, 4A, 01, 14, 6A, 2A, 0A, 07}
NXP_RFEXT_CONFIG_2={20, 02, 0C, 01, A2, 0F, 08, 00, FF, 01, 01, 32, 28, 13, 00}
NXP_RFEXT_CONFIG_3={20, 02, 0C, 01, A2, 10, 08, 00, 28, 13, 00, 00, 28, 13, 00}
NXP_RFEXT_CONFIG_4={20, 02, 0C, 01, A2, 11, 08, 00, 26, 13, 00, 00, 23, 13, 00}
NXP_RFEXT_CONFIG_5={20, 02, 0C, 01, A2, 12, 08, 00, 21, 13, 00, 00, 20, 00, 00}
NXP_RFEXT_CONFIG_6={20, 02, 0C, 01, A2, 13, 08, 00, 1F, 00, 00, 00, 1E, 00, 00}
NXP_RFEXT_CONFIG_7={20, 02, 0C, 01, A2, 14, 08, 00, 1D, 00, 00, 00, 1D, 00, 00}
NXP_RFEXT_CONFIG_8={20, 02, 0C, 01, A2, 15, 08, 00, 1D, 00, 00, 00, 1C, 00, 00}
NXP_RFEXT_CONFIG_9={20, 02, 0C, 01, A2, 16, 08, 00, 1C, 00, 00, 00, 1B, 00, 00}
NXP_RFEXT_CONFIG_10={20, 02, 0C, 01, A2, 17, 08, 00, 1B, 00, 00, 00, 1A, 00, 00}
NXP_RFEXT_CONFIG_11={20, 02, 0C, 01, A2, 18, 08, 00, 1A, 00, 00, 00, 19, 00, 00}
NXP_RFEXT_CONFIG_12={20, 02, 0C, 01, A2, 19, 08, 00, 19, 00, 00, 00, 18, 00, 00}
NXP_RFEXT_CONFIG_13={20, 02, 0C, 01, A2, 1A, 08, 00, 18, 00, 00, 00, 17, 00, 00}
NXP_RFEXT_CONFIG_14={20, 02, 0C, 01, A2, 1B, 08, 00, 17, 00, 00, 00, 16, 00, 00}
NXP_RFEXT_CONFIG_15={20, 02, 0C, 01, A2, 1C, 08, 00, 16, 00, 00, 00, 16, 00, 00}
NXP_RFEXT_CONFIG_16={20, 02, 0C, 01, A2, 1D, 08, 00, 15, 00, 00, 00, 15, 00, 00}
NXP_RFEXT_CONFIG_17={20, 02, 0C, 01, A2, 1E, 08, 00, 14, 00, 00, 00, 13, 00, 00}
NXP_RFEXT_CONFIG_18={20, 02, 0C, 01, A2, 1F, 08, 00, 13, 00, 00, 00, 12, 00, 00}
NXP_RFEXT_CONFIG_19={20, 02, 0C, 01, A2, 20, 08, 00, 12, 00, 00, 00, 12, 00, 00}
NXP_RFEXT_CONFIG_20={20, 02, 0C, 01, A2, 21, 08, 00, 11, 00, 00, 00, 11, 00, 00}
NXP_RFEXT_CONFIG_21={20, 02, 0C, 01, A2, 22, 08, 00, 10, 00, 00, 00, 10, 00, 00}
NXP_RFEXT_CONFIG_22={20, 02, 0C, 01, A2, 23, 08, 00, 0F, 00, 00, 00, 0E, 00, 00}
NXP_RFEXT_CONFIG_23={20, 02, 0C, 01, A2, 24, 08, 00, 0C, 00, 00, 00, 0A, 00, 00}

# ProtocolArea Settings
#####

#####
```

#### 10. 执行脚本

**C:\nxp\NxpNfcCockpit\_v8.3.0.0\firmware\Secondary\_Pn722x\Android\restore\_NFC\_HAL.bat**恢复hal库

**注意：不执行该脚本的话nfc功能无法打开**

#### 11. 将该文件push到设备中重启验证nfc功能

```
adb root
adb remount
adb push libnfc-nxp-rfExt.conf /vendor/etc/libnfc-nxp-rfExt.conf
adb reboot
adb install --bypass-low-target-sdk-block "NFC Tools Pro v1.8 已付費繁體中文版.apk"
```

#### 12. 读卡功能ok

